Petualangan Ajisaka - Technical Documentation

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Contents

1 Petualangan Ajisaka — Aplikasi Pembelajaran Menulis Aksara Jawa

Design & Development Document

Version	0.1 (draft)
Date	2026-08-12
Status	Design document — pre-build
Delivery target	Installable Progressive Web App (PWA)
Source material	`projek Isif.pdf` — *Naskah Lengkap Aplikasi Petualangan Ajisaka*

1.1 1. Overview

Petualangan Ajisaka is a gamified learning application that teaches children to read and write **Aksara Jawa** (the Javanese script) through a story-driven adventure. The player follows the legendary hero **Ajisaka** across three islands, unlocking relics and defeating enemies by completing on-screen handwriting practice challenges.

The app runs entirely inside the browser and is packaged as an installable, **offline-capable PWA** so students on low-end tablets and school Chromebooks can play without connectivity.

1.1.1 1.1 What the product is

- A *stroke-practice* game: the player literally writes Aksara Jawa with a finger or stylus on a canvas, guided by animated arrow guides (stroke order + direction).
- A *narrative adventure*: writing missions are framed as story events (unlock a sword seal, answer a village elder's test, battle a giant's envoys).
- A *progression system*: three levels (Pemula → Mahir → Master), rewards (Pedang Pusaka, Perisai Sakti), followers joining the party (Dora, the local villager), and a final exam + crowned-ending.

1.2 2. Goals & Non-Goals

1.2.1 2.1 Goals

#	Goal
G1	Teach correct **stroke order and direction** for Aksara Jawa basics (Nglegena), Sandangan, and Pasangan.
G2	Keep learners motivated through narrative, rewards, and escalating difficulty.
G3	Work **offline** and be **installable** on Android tablets, iPads, and desktop browsers (PWA).
G4	Give immediate, friendly feedback on each stroke (truthy/dirty, live).
G5	Track progress locally per device (no account required for MVP).
G6	Provide a complete, reusable dataset of Aksara Jawa strokes and Unicode mappings for future features (transliteration, quizzes).

1.2.2 2.2 Non-goals (MVP)

- Server accounts, cloud sync, or leaderboards (future work).
- Freehand recognition beyond the taught question set (no arbitrary text input).
- Audio instructions / speech synthesis (future; on-screen text only).
- Multi-language UI beyond Indonesian + minimal English (future).
- Native app stores (PWA only for now; TWA/APK wrapper considered later).

1.3 3. Target Users & Personas

Persona	Profile	Needs
Adi (9 y.o.)	Primary: elementary student learning Aksara Jawa in class	Playful, guided practice, immediate feedback, feels like a game
Bu Sari (32 y.o.)	Teacher	Assignable practice, clear level map, zero-install on classroom tablets
Pak Damar (40 y.o.)	Parent helping at home	Simple install, offline, no account, safe kid UI

Design consequence: big touch targets (48px), high-contrast solid buttons, readable Indonesian copy, minimal typing, zero ads, zero tracking.

1.4 4. Narrative & Content Spec (from source PDF)

1.4.1 4.1 High-level arc

`Petualangan Ajisaka`

```
Prolog  Mengenal Asal-usul Aksara Jawa (edukasi)
Level 1  Pemula   · Pulau Sanjaya    → Pedang Pusaka    (+ Dora joins)
Level 2  Mahir    · Pulau Adi Jaya   → Perisai Sakti    (+ warga lokal joins)
Level 3  Master   · Kerajaan Nusantara (2 fase) → Raksasa Hijau defeated → crowned King
```

1.4.2 4.2 Screen / menu breakdown

#	Screen	Content	Writing mechanic
0	**Halaman Awal (Splash/Home)**	Media title + big **PLAY** button	—
1	**Dashboard (Main Menu)**	5 navigation menus: Prolog, Level 1, Level 2, Level 3, *Kembali ke Rumah*	—
2	**Prolog**	Educational page: history & origin of Aksara Jawa	—
3	**Level 1 — Pemula** (Pulau Sanjaya)	Sinopsis: unlock seal + take the sword. **Narasi sinopsis tidak diekstrak utuh dari PDF →** open question §20	Write **Aksara Dasar (Nglegena)** with **arrow stroke-guides**
4	**Level 1 End**	Reward: **Pedang Pusaka**; meeting **Dora** joins the party	—
5	**Level 2 — Mahir** (Pulau Adi Jaya)	Sinopsis: sail with Dora, find **Perisai Sakti**, a local villager sets a test	**11 soal** (per narrative) of Sandangan writing
6	**Level 2 End**	Reward: **Perisai Sakti**; villager joins party; *Next Level*	—
7	**Level 3 — Master, Fase 1** (Penghadangan Dua Utusan)	Two Green Giant envoys intercept the ship at sea; defeat them by writing	**20 soal Aksara Pasangan** (stacked consonants: main letter + attached pasangan below/side), full stroke-guide
8	**Level 3 — Master, Fase 2** (Penyegelan Raksasa Hijau)	Infiltrate Nusantara Kingdom; free it from the Green Giant	**3 latihan menus × 5 soal = 15 soal**: (1) kalimat dgn Aksara Dasar, (2) kalimat dgn Sandangan, (3) kalimat dgn Pasangan
9	**Akhir Cerita**	Happy ending: Green Giant sealed away; player crowned **Raja Kerajaan Nusantara**	—

1.4.3 4.3 Numbers locked from the narrative

Item	Count	Source
Level 2 writing questions	**11**	"ke-11 soal"
Level 3 Fase 1 questions	**20** (pasangan)	narrative
Level 3 Fase 2 questions	**15** (3×5)	narrative
Level 1 question count	*unspecified*	**open question §20**

1.4.4 4.4 Content gaps to confirm (from PDF anomalies)

1. Level 1 synopsis text for Pulau Sanjaya (the "unlock the seal" beat is present, but the full story paragraph was empty in extraction).
2. Whether Level 1 *requires* Nglegena only, or includes a tutorial sub-mode.
3. Reward naming convention + visual style (game-asset source or in-app SVG?).

1.5 5. Functional Requirements

1.5.1 FR-1 · Home & Navigation

- **FR-1.1** Home screen shows the media title and a primary **PLAY** button.
- **FR-1.2** PLAY opens the dashboard; system back or "Kembali ke Rumah" returns home.
- **FR-1.3** Dashboard always lists `Prolog`, `Level 1`, `Level 2`, `Level 3`, `Kembali ke Rumah`.
- **FR-1.4** Locked levels are visually locked and show a "complete previous level" hint (do not hard-block navigation; allow reading Prolog anytime).

1.5.2 FR-2 · Story / Prolog

- **FR-2.1** Prolog renders the origin story of Aksara Jawa as illustrated scenes.
- **FR-2.2** Supports tap-to-advance (like a slide deck); skippable with confirmation.

1.5.3 FR-3 · Writing Practice (core gameplay)

- **FR-3.1** Each question shows a reference glyph (rendered with the Javanese font) + its romanization + Indonesian hint.
- **FR-3.2** A canvas area captures freehand strokes (single touch/stylus; multi-touch disabled).
- **FR-3.3** Animated **stroke guide**: numbered arrow heads showing direction and order, replayable on demand.
- **FR-3.4** Live feedback: stroke match indicator (e.g., green = good, amber = acceptable, red = wrong direction/order).
- **FR-3.5** Between strokes, guide for the next sub-stroke highlights; completed strokes stay visible.
- **FR-3.6** **Undo / Clear** controls (big, kid-safe).
- **FR-3.7** After a question completes: celebratory micro-feedback, then auto-advance or "Lanjut".
- **FR-3.8** A wrong-but-attempted answer can be retried (no fail-state lock on MVP).

1.5.4 FR-4 · Level Flow & Rewards

- **FR-4.1** Level intro screen = story beat (sinopsis) → practice session → outro beat.

- **FR-4.2** On completion show reward item + story update (e.g., "Dora bergabung!").
- **FR-4.3** "Next Level" button unlocks the following level.
- **FR-4.4** Level 3 has two phases with a phase-transition interstitial.

1.5.5 FR-5 · Persistence

- **FR-5.1** Persist: completed levels, per-question best score, collected rewards, settings.
- **FR-5.2** Resumable: closing mid-level returns to dashboard with progress intact; continue begins from first unanswered question.
- **FR-5.3** Storage: local-first (IndexedDB via `localStorage` facade for MVP state; canvas data ephemeral).

1.5.6 FR-6 · PWA

- **FR-6.1** Installable (valid manifest + icons + service worker).
- **FR-6.2** Official **offline-first**: full app shell + asset (fonts, data, icons) precache; no network needed to play.
- **FR-6.3** Updates: versioned SW with prompt-to-refresh when a new release is cached.

1.6 6. Information Architecture & Navigation

```
Home (Halaman Awal)
    [PLAY]

Dashboard  (5 menu)
   Prolog  Story slides  back
   Level 1 (Pemula)  [Intro sinopsis]  Practice L1  [Reward: Pedang]  Next Level
   Level 2 (Mahir)  [Intro sinopsis]  Practice L2 (11)  [Reward: Perisai]  Next Level
   Level 3 (Master)  Fase 1: Practice Pasangan (20)  Fase 2: Lat.1 (5) Lat.2 (5) Lat.3 (5)  Enc
   Kembali ke Rumah  Home
```

1.6.1 6.1 Navigational rules

- Global, persistent **back** affordance on every non-dashboard screen (system back + visible button).
- Dashboard is the central hub; all level screens return to it.
- *Next Level* is the only forward gate; it appears only after a level completes.

1.7 7. User Flows

1.7.1 7.1 First play through Level 1

```
flowchart TD
  A[Home] -->|PLAY| B[Dashboard]
  B -->|Level 1| C["Intro sinopsis: Pulau Sanjaya"]
  C --> D["Writing practice: Nglegena set"]
  D -->|question done| D
```

```
  D -->|all done| E["Outro: Pedang Pusaka + Dora joins"]
  E -->|Next Level| F[Dashboard - Level 2 unlocked]
```

1.7.2 7.2 Single-question writing loop

```
flowchart TD
  A[Show glyph + hint] --> B[Replay stroke guide]
  B --> C[Player traces on canvas]
  C --> D{Stroke match engine}
  D -->|pass| E["Stroke locked green - next guide"]
  D -->|dirty - retry stroke| C
  E --> F{All sub-strokes done?}
  F -->|no| B
  F -->|yes| G["Celebration - record score"]
  G --> H{More questions?}
  H -->|yes| A
  H -->|no| I[Level outro + reward]
```

1.8 8. Screen Inventory (route map)

Route	Screen	Key UI
`/`	Home	Title, PLAY
`/menu`	Dashboard	5 level cards
`/prolog`	Prolog story	Slides, skip
`/level/1`	L1 flow	Intro / practice / outro
`/level/2`	L2 flow	Intro / practice(11) / outro
`/level/3`	L3 flow	Fase1 practice(20) → Fase2 3×latihan(15) → ending
`/ending`	Happy ending	Crown ceremony

(Routes are client-side only; PWA serves the app shell for every path → use hash router or SW navigation fallback to `/`.)

1.9 9. Design System

1.9.1 9.1 Design language

Three layers (per product guardrails):

1. **Material You** — interaction model: adaptive touch surfaces, tonal surfaces, dynamic elevation, large padded buttons, rounded corners.
2. **Minimalism** — layout discipline: generous whitespace, clear hierarchy, one primary action per screen, reduced visual noise so a 9-year-old isn't overwhelmed.

3. **Glassmorphism** — only as an *accent* on story/interstitial screens (frosted panels over illustrated backgrounds); not on gameplay surfaces (sketch canvases must stay glare-free for writing).

Genre: **playful** (post-Linear soft school: consumer, casual, children). Warm, pastel-dominant palette with deep "keraton" accents. Illustrations are the brand: hand-drawn Ajisaka line-art silhouettes (simple SVG), not photos.

1.9.2 9.2 Palette (Keraton theme)

Derived from the playful **Carnival** token system, re-hued toward Javanese heritage colors (keraton indigo/red-gold + batik neutrals), pastel-first.

Token	Value (OKLCH)	Use
`--color-paper`	`oklch(97% 0.012 78)`	App background (batik cream)
`--color-paper-2`	`oklch(92% 0.03 78)`	Card surfaces
`--color-paper-3`	`oklch(84% 0.05 55)`	Hover/interactive surface
`--color-text`	`oklch(34% 0.04 262)`	Primary text (deep indigo-black)
`--color-text-2`	`oklch(52% 0.03 262)`	Muted text
`--color-accent`	`oklch(58% 0.14 25)`	Primary action (keraton red / terracotta)
`--color-accent-2`	`oklch(70% 0.13 80)`	Success / gold (pedang & perisai rewards)
`--color-warn`	`oklch(66% 0.14 60)`	Warning / retry stroke
`--color-error`	`oklch(52% 0.16 27)`	Error / wrong stroke
`--color-border`	`oklch(86% 0.025 78)`	Dividers
`--color-focus`	`oklch(60% 0.16 262)`	`:focus-visible` ring (indigo)
`--color-glass`	`oklch(100% 0 0 / 0.55)`	Frosted overlay fill

Dark mode: derived automatically by inverting paper band (keep for classroom) via `@media (prefers-color-scheme: dark)` + `[data-theme=dark]`.

1.9.3 9.3 Typography

Role	Token	Stack
Display / game titles	`--font-display`	`'Fredoka One', 'Baloo 2', system-ui, sans-serif` (rounded, kid-friendly)
Body / copy	`--font-body`	`'Nunito', system-ui, sans-serif`
Code / glyph data labels	`--font-mono`	`'JetBrains Mono', monospace`

Scale (from token system): display `clamp(2.5rem,6vw,4.5rem)`, display-s `clamp(2rem,4vw,3rem)`, 2xl `1.75rem`, xl `1.25rem`, lg `1.125rem`, base `1rem`, sm `0.875rem`, xs `0.75rem`.

Rules: headings always roman (no italic headings); italics only inside body copy for emphasis. Indonesian copy throughout; keep sentences short (12 words).

Javanese glyph font: self-hosted **Noto Sans Javanese** (Unicode block U+A980–U+A9DF) for rendering reference glyphs. Fallback font-stack on canvas: `'Noto Sans Javanese', sans-serif`. (Earliest glyph loads are precached by the SW.)

1.9.4 9.4 Spacing, radius, elevation

- Spacing: 4px base scale (xs 4, sm 8, md 12, lg 16, xl 24, 2xl 32, 3xl 48 . . .).
- Radius: sm 4 / md 8 / lg 16 — reuse, never invent.
- Elevation: Material tonal elevation — cards = paper-2 on paper; modals use `box-shadow` with glass blur (story interstitials only).
- Touch targets: **min 48×48px**, interactive gap 8px.

1.9.5 9.5 Motion

- Durations: fast 150ms (hover/micro), base 250ms, slow 400ms (screen transitions).
- Easings: `cubic-bezier(0.16,1,0.3,1)` enter / `cubic-bezier(0.4,0,0.68,0.06)` exit.
- All entrance/stagger/counter/loader animation through **anime.js v4** snippets, each guarded by `prefers-reduced-motion`.
- Story slides: crossfade + slight scale; gameplay feedback: quick spring on stroke-lock.
- **Game feel:** confetti-ish particle burst on question completion and at level rewards (Web Canvas, keep under 60 particles).

1.9.6 9.6 Iconography

Font Awesome (free) per defaults; considered alternatives: Material Symbols, Lucide. Use spellings/visuals a child recognizes (shield, sword, home, arrow, question, star). All icon buttons carry `aria-label`.

1.9.7 9.7 Components (8-state discipline)

Component	States to style
`Button` (primary/ghost/danger)	default, hover, focus-visible, active, disabled, loading, error, success
`LevelCard`	default, hover, focus, active, locked, completed, current, disabled
`PracticeCanvas`	idle, guiding, drawing, stroke-locked, feedback(ok/warn/error), cleared
`IconButton` (back/undo/clear/replay)	all 8 states
`Modal` (congrats, reset, update)	open/close anim + focus trap

1.10 10. Domain Model & Data

1.10.1 10.1 Aksara dataset (`src/data/aksara.ts`)

```
type AksaraType = 'nglegena' | 'sandangan' | 'pasangan';

interface AksaraGlyph {
  id: string;                    // "ha", "na", "ca" ...
  type: AksaraType;
  label: string;                 // romanization
  hintID: string;                // Indonesian hint translation key
  unicode: string;               // e.g. "\uA98F" (ha)
  unicodePasangan?: string;      // pasangan glyph codepoint where applicable
  /** Normalized vector strokes, in units of the write-box (0..1). */
  strokes: Stroke[];             // ordered! = stroke-order/isomorphism
  sound?: string;                // phoneme for future TTS
}

interface Stroke {
  points: Array<{ x: number; y: number }>;  // polyline, 0..1 box coords
  tolerance: number;                         // per-stroke looseness (default 0.12)
}
```

1.10.2 10.2 Level config

```
interface LevelConfig {
  id: number;
  title: string;          // "Level 1 · Pemula · Pulau Sanjaya"
  questions: Question[];
  reward: { id: string; name: string };   // "Pedang Pusaka"
  story: { intro: string; outro: string };
}

interface Question {
  id: string;
  glyphId: string;        // → AksaraGlyph
  prompt: string;         // "Tulis aksara: HA (dasar)"
  sentence?: string;      // full Javanese sentence for Fase-2 questions
}
```

1.10.3 10.3 Seed content (proposed; volumes = open questions)

Set	Type	Count
Nglegena (dasar)	20 + optional variants	used in L1, L3-Fase2 menu 1
Sandangan	8 core (a→i,u,e,o, taling/tarung, pepet, cecak, etc.)	L2 (11 soal incl. combos), L3-F2 menu 2
Pasangan	pair of main + pasangan glyph	L3-F1 (20 soal), L3-F2 menu 3

Final counts per level confirmed in §4.3; question contents (which glyphs) are a **curriculum decision** — propose defaults §20.

1.11 11. Stroke Engine Design (core challenge)

1.11.1 11.1 Capture

- `<canvas>` full viewport of the write-box; **Pointer Events** (`pointerdown/move/up`) unified for touch + stylus + mouse; `touch-action: none` to stop scrolling.
- Points downsampled (10px spacing) and normalized into the 0..1 write-box.
- HiDPI: canvas sized to `devicePixelRatio`, CSS box fixed.

1.11.2 11.2 Stroke-guide rendering

- Draw each reference `Stroke` polyline as faint target guide.
- Animated progress along the path = the classic "arrow head" tracer (anime.js or requestAnimationFrame), replayable via **Replay** button.
- Next expected sub-stroke highlighted; completed strokes rendered solid + check.

1.11.3 11.3 Online matching (offline, deterministic, fast)

Approach to keep 0 weight and run on-device: 1. **Normalize** the input stroke (resample to N=32 points, scale to 0..1, translate to center, keep proportional aspect). 2. **Direction signature:** per-segment angle buckets (e.g., 8 bins). 3. **Dynamic Time Warping (DTW)** distance to the reference normalized stroke (proxy: use multi-step/rough DTW on resampled polylines). 4. **Order = truth:** strokes must be completed in the reference order; a reversal triggers the "wrong direction" feedback even if the shape matches. 5. Score → `pass / warn / retry` via per-stroke tolerance.

Optional future upgrade: a tiny **ONNX/TFLite** model (`12 class output`, < 500 KB) trained on augmentations of the 32 reference glyphs, run via `@xenova/transformers` or plain TensorFlow.js — swapped behind the same `StrokeMatcher` interface. **Not in MVP.**

1.11.4 11.4 Feedback mapping

Signal	Meaning	Color
pass	stroke matched shape+order	green (`accent-2`)
warn	shape ok, slight drift	gold (`warn`)
retry	wrong direction / order	red (`error`)

1.12 12. PWA & Architecture

1.12.1 12.1 Recommended stack

Concern	Choice
Language	**TypeScript (strict)**

Concern	Choice
Build	**Vite**
UI framework	**React 18** (or Svelte if a lighter runtime is preferred — see §20)
Styling	**Tailwind CSS v4** with CSS custom-property tokens (OKLCH)
State	Zustand with `persist` middleware (localStorage)
Routing	`react-router` + **hash routing** (offline-friendly, no server rewrites needed)
PWA	`vite-plugin-pwa` (Workbox): manifest + SW precache + offline fallback
Canvas/game	Native Canvas 2D + anime.js v4 (motion)
Icons	Font Awesome (free)
Data-local	JSON/TS dataset (§10) + `localStorage` for progress
Tests	Vitest + Testing Library; Playwright (E2E, offline simulate)
Lint/format	ESLint (flat) + Prettier (matches existing `.pre-commit-config.yaml`)
Icons/app assets	inline SVG (kirik) so SW precache stays small

1.12.2 12.2 Layered architecture

```
UI Layer        React components, routes,
                screens, story slides

Application     LevelFSM, Question session,
                progress store, reward logic

Domain          aksara.ts dataset, Stroke
                matcher interface, scoring

Infrastructure  CanvasEngine (capture+guide),
                DTW matcher, Zustand persist,
                PWA/SW, audio (gamelan blips)
```

1.12.3 12.3 State model (Zustand useProgress)

```
interface ProgressState {
  completedLevels: number[];          // [1,2,3]
  rewards: string[];                  // ['pedang','perisai']
  bestByQuestion: Record<string, number>; // questionId -> percent score
  currentLevel?: number;              // resume point
  unfinished: string[];               // unresolved question ids in current level
```

```
  settings: { sound: boolean; dark: boolean };
}
```

1.12.4 12.4 Offline & SW strategy

- **Precache (build-time, hashed):** HTML shell, JS/CSS chunks, Javanese font (woff2), dataset, inline SVG icons — the entire app works with network off.
- **Runtime cache:** none required for MVP (all local); precache covers everything.
- **Update flow:** `registerSW` + prompt-to-refresh banner when `updatedready`.
- **No backend:** zero API calls; CSP tightened; no analytics.
- Manifest: name “Petualangan Ajisaka”, `display: standalone`, `orientation: portrait` (writing optimised), theme color = `--color-paper`, icon set incl. `512px` maskable.

1.13 13. Proposed Project Structure

```
javanese_learning_app/
 public/
   icons/            (192, 512, maskable)
   fonts/            (NotoSansJavanese woff2)
 src/
   data/             aksara.ts, levels.ts, sentences.ts
   engine/           capture.ts, normalize.ts, dtw.ts, matcher.ts
   state/            progress.ts
   ui/               components/, screens/
   hooks/            useCanvas.ts, useStrokeSession.ts
   styles/           tokens.css, base.css
   App.tsx
   main.tsx
 vite.config.ts        (+ vite-plugin-pwa)
 tsconfig.json
 index.html
 DESIGN_AND_DEV.md
```

1.14 14. Testing Strategy

- **Unit:** `dtw.ts` (match/fail cases), `normalize.ts`, progress-store transitions, question scoring, level-unlock gating.
- **Component:** practice screen states (empty/guiding/warn/error/success), canvas button labels, modal focus trap.
- **E2E (Playwright):** flow Home→L1 completed→L2 unlocked; reload mid-level resumes; **offline emulation** verifies full playthrough with no network.
- **Manual device matrix:** iPad, Android tablet (Chrome), desktop (portrait mobile view).
- **A11y:** axe-core scan in CI.
- Pre-commit (existing hooks) keep running; add `eslint/prettier/tsc --noEmit` there.

1.15 15. Performance Budget (offline-first)

Metric	Budget
First load (offline cache hit)	1.5 MB JS + assets; font 600 KB woff2
TTI on mid-range tablet	3 s
SW precache total	4–5 MB
Canvas frame during tracing	60 fps
Recognition latency	< 50 ms (DTW on 32-pt resample)

1.16 16. Accessibility (WCAG 2.1 AA)

- Color is never the only signal: pass/warn/error carry icons + text.
- Focus-visible ring `--color-focus`, `outline-offset: 2px`; skip-link; landmark tags.
- Keyboard operable (arrows on story slides; Enter/Space on controls); canvas has an accessible fallback description for each question.
- `prefers-reduced-motion` disables stroke-guide animation, particles, slide fx.
- Contrast: body text 4.5:1; UI 3:1 (verify with WebAIM checker on final hex).
- All Indonesian copy short, plain; alt text on illustrated scenes.

1.17 17. Security

- No user input is rendered as HTML (all content from bundled dataset).
- CSP: `default-src 'self'; script-src 'self'; font-src 'self'; img-src 'self' data:;` (no external CDNs at runtime — everything precached).
- No secrets/keys in client (none needed); no network traffic to leak.
- Canvas input validated/ignored outside the write-box; multi-touch suppresses stray strokes.

1.18 18. Roadmap / Milestones

Milestone	Scope
M0 — Seed & shell	Vite+TS app, tokens, routes, dashboard, manifest/SW, empty practice screen
M1 — Stroke engine	Canvas capture, guide animation, DTW matcher, feedback mapping
M2 — Content	Full dataset (Nglegena 20+ / Sandangan 8+ / Pasangan pairs), level configs, story copy in Indonesian
M3 — Game flow	Level FSM, rewards, Dora/villager joins, ending, progress persistence
M4 — Polish QA	Particles, sound, offline E2E, a11y scan, device matrix, update banner

---

1.19 19. Future Work (after MVP)

- Transliteration / text-to-Aksara input; pronunciation (gamelan + TTS).
- Cloud sync & teacher analytics (rename: "Guru" report).
- Android APK via TWA; wider glyph set (Angka, Murda, Swara).
- ML stroke classifier upgrade behind the matcher interface.

1.20 20. Open Questions & Assumptions

#	Question	Proposed default / status
1	Level 1 question count	Propose **10–12** Nglegena; confirm with curriculum.
2	Exact question contents per set	Propose starter sets; finalize with teacher input.
3	Full Level-1 sinopsis paragraph	Missing from PDF extraction — supply copy.
4	Reward/asset art source	Inline SVG illustrations; replaceable later.
5	UI framework	Recommend **React**; Svelte alternative if bundle size is critical.
6	Portrait-only PWA OK?	Yes (writing orientation); desktop shows rotated mock.

1.21 21. References

- Source narrative: `projek Isif.pdf` (Naskah Lengkap Aplikasi Petualangan Ajisaka).
- Font: Noto Sans Javanese (OFL) — self-hosted.
- Standards: WCAG 2.1 AA; PWA installability criteria (Chrome/Android, iOS 16+).
- Design system basis: playful *Carnival* token set (paper/accent OKLCH architecture) re-hued to the keraton palette in §9.2.

End of document. Next step: confirm §20 decisions, then scaffold M0.# Naskah Aplikasi Petualangan Ajisaka

Tema utama: gamifikasi keterampilan menulis Aksara Jawa.

1.22 Halaman Utama dan Navigasi

Aplikasi dibuka dengan halaman depan yang menampilkan judul media pembelajaran dan satu tombol PLAY besar. Setelah tombol itu ditekan, pemain masuk ke menu utama yang menampung lima pilihan:

1. **Prolog** — pengenalan cerita dan asal-usul.
2. **Level 1: Pemula** — menulis aksara dasar.
3. **Level 2: Mahir** — menulis aksara sandangan.
4. **Level 3: Master** — menulis aksara pasangan dan ujian akhir.
5. **Kembali ke Rumah** — kembali ke halaman depan.

1.23 Prolog: Mengenal Asal-Usul

Pilihan pertama membawa pemain ke halaman edukasi yang membahas sejarah terciptanya Aksara Jawa. Bagian ini menjawab dari mana aksara itu lahir dan bagaimana ia sampai dipakai seperti sekarang, sebelum pemain mulai menulis satu huruf pun.

1.24 Level 1 — Pemula: Misi di Pulau Sanjaya

Di pulau pertama, Ajisaka harus membuka segel dan mengambil pedang. Syaratnya jelas: pemain wajib menyelesaikan tantangan praktik menulis Aksara Jawa Dasar (Nglegena) langsung di layar digital. Proses penulisannya dibantu penuh oleh garis panduan berbentuk anak panah, atau *stroke guide*, yang menunjukkan arah guratan tangan secara tepat — dari coretan pertama sampai hurufnya jadi.

Setelah tantangan menulis selesai, pemain menerima hadiah berupa **Pedang Pusaka**. Tak lama berselang, Ajisaka bertemu seorang pemuda lokal bernama Dora yang memutuskan menjadi pengikut setianya. Ketika cerita selesai, muncul tombol **Next Level** untuk melanjutkan ke menu ketiga.

1.25 Level 2 — Mahir: Misi di Pulau Adi Jaya

Perjalanan berlanjut. Ajisaka dan Dora mengarungi pulau kedua, Pulau Adi Jaya, untuk mencari pusaka berikutnya: **Perisai Sakti**. Di tengah jalan mereka bertemu seorang warga lokal yang mengetahui tempat penyimpanan perisai itu. Warga tersebut menantang mereka, berjanji kembali memberi tahu lokasinya bila Ajisaka dan Dora mampu menjawab ujian yang ia berikan.

Melihat kemahiran Ajisaka menuliskan ke-11 soal ujian itu, sang warga kagum dan berniat ikut bertualang melawan raksasa. Ia menunjukkan letak Perisai Sakti hingga pusaka itu berhasil diambil. Kini mereka bertiga bersiap berlayar ke pulau utama lewat tombol **Next Level**.

1.26 Level 3 — Master: Pertempuran Akhir di Kerajaan Nusantara

Tingkat terakhir ini terbagi menjadi dua fase pertempuran menulis yang sengit.

1.26.1 Fase 1: Penghadangan Dua Utusan

Dalam perjalanan laut menuju pusat kerajaan musuh, kapal mereka dicegat dua utusan Raksasa Hijau yang berniat membunuh Ajisaka. Pemain wajib menyelesaikan 20 soal Aksara Pasangan, yaitu konsonan tumpuk. Pada tiap soal, pemain diminta menuliskan Aksara Jawa utama sekaligus bentuk pasangannya yang menempel di bawah atau di samping. Seluruh guratan dibimbing arah anak panah agar susunan penulisannya tidak keliru. Begitu ke-20 soal selesai ditulis, kedua utusan itu berhasil diselesaikan.

1.26.2 Fase 2: Ujian Penyegelan Raksasa Hijau

Sistem langsung mengoper pemain ke menu latihan soal akhir. Ajisaka, Dora, dan warga lokal itu berhasil menyusup ke Kerajaan Nusantara yang sedang dikuasai Raksasa Hijau. Untuk memerdekakan kerajaan dari teror, pemain harus menyelesaikan tiga menu latihan utama; masing-masing berisi lima soal menulis:

1. **Latihan 1** — menulis kalimat bahasa Jawa menggunakan komponen Aksara Jawa Dasar.
2. **Latihan 2** — menulis kalimat bahasa Jawa yang menggunakan aturan Aksara Sandangan.
3. **Latihan 3** — menulis kalimat bahasa Jawa yang menggunakan aturan Aksara Pasangan.

1.26.3 Akhir Cerita

Setelah ketiga menu latihan penulisan, total 15 soal, diselesaikan secara berurutan, Raksasa Hijau berhasil diselesaikan secara mutlak. Kerajaan Nusantara kembali damai dan bebas dari penjajahan. Atas jasa dan kepemimpinannya, pemain yang memerankan Ajisaka secara resmi dinobatkan menjadi **Raja Kerajaan Nusantara**.# Petualangan Ajisaka — Todo Pengembangan (P0–P3)

Prioritas berdasarkan `DESIGN_AND_DEV.md` §5 & `TIMELINE.md` W1–W11. *Diperbarui setelah M0+M1 PoC build (2026-08-12).*

Skala prioritas - **P0 — Kritis:** tanpa ini aplikasi tidak bisa rilis / tidak bisa dimainkan. - **P1 — Penting:** wajib untuk peluncuran yang baik, dapat ditunda jika terdesak. - **P2 — Peningkatan:** nilai tambah setelah rilis inti stabil. - **P3 — Masa depan:** di luar scop MVP, direncanakan kemudian.

- ☒ = selesai & terverifikasi (build/lint/test hijau)
- [~] = sebagian / versi PoC / butuh kurasi

1.27 P0 — Kritis (rintisan rilis)

1.27.1 Fondasi & shell — DONE (M0)

- ☒ Scaffold Vite + TypeScript strict + Tailwind v4 — `src/`
- ☒ Token desain keraton (OKLCH) §9.2 — `src/src/index.css` (@theme)
- ☒ Routing hash: `/`, `/menu`, `/prolog`, `/level/1..3`, `/ending` — App.tsx
- ☒ Halaman Home (judul + tombol PLAY) — FR-1.1 (Redesigned with playful UX)
- ☒ Dashboard 5 menu dengan state locked/unlock — FR-1.3, FR-1.4 (Redesigned with Gamified LevelCards)
- ☒ Manifest PWA + icons (192/512/maskable/apple-touch) + installable
- ☐ Finalisasi keputusan §20 (jumlah soal L1, isi dataset, framework React, aset ilustrasi) — sebagian tetap terbuka

1.27.2 Stroke engine — PIVOT: DTW → VECTOR GEOMETRY

- ☒ Hook canvas Pointer Events, HiDPI, `touch-action: none`, live-ink — useCanvasCapture

- ☒ Normalisasi + resample (engine/geometry) — dipakai rendering
- ☒ Matcher **Vector Geometry (Chamfer Dist)** — engine/raster.ts (Skala dinamis, cegah exploit coretan)
- ☒ Uji matcher sintetis (square/fill/miss/offset) + 20 glyph contours — vitest 8 tests
- ☒ Umpan balik warna per guratan (green/amber/red) — §11.4
- ☒ Kontrol Clear ("Bersihkan") — FR-3.6
- ☒ Kalibrasi toleransi coverage (COVERAGE_PASS/WARN) dgn uji tablet nyata — sudah dinamis per luas/tebal glyph

1.27.3 Konten

- ☒ Dataset Nglegena **20 contour** (dari Noto Sans Javanese, Y-corrected, 80-pt) — src/data/nglegena_contours.json
- ☒ Dataset Sandangan 8+ core + kombinasi — §10.1 (Label & Dotted Circle diperbaiki)
- ☒ Dataset Pasangan (20 pasang) — §10.1 (Berhasil diekstrak ulang penuh dengan loops via FontTools Python script)
- ☒ Level config L1–L3 bukti: starter ha/na/ca/ra/ka; soal 11/20/3×5 belum terisi penuh — §10.2 (Dataset penuh digunakan untuk L1-L3)
- ☒ Copy cerita Indonesia penuh (dari `NASKAH_AJISAKA.md`) ke tiap level — Selesai via i18n layer
- ☒ Precache font Javanese (woff2, 89KB, self-host) — FR-6.2
- ☐ **Stroke data kurasi internal (guide pen-stroke ori)** — kanvas guide saat ini solid-fill contour; untuk guratan internal/order diajarkan butuh data kurator (§20)

1.27.4 Game flow & state

- ☒ Level FSM path: intro screen → practice → done page
- ☒ Reward Pedang Pusaka (L1) / Perisai Sakti (L2) / Raja (L3) — LevelDone
- ☒ Event "Dora bergabung" & "warga bergabung" (copy) — LevelDone
- ☒ Unlock chain + tombol Next Level — FR-4.3
- ☒ Interstitial Fase 1 / Fase 2 pilihan menu non-linear di Level 3 — FR-4.4 (Selesai, tracking granular per phase)
- ☒ Ending dinobatkan Raja terhubung setelah L3 Fase 1 & 2 selesai — Selesai
- ☒ Persist progress (Zustand persist): completedLevels, completedPhases, rewards — FR-5.1
- ☐ Resume level saat reload — FR-5.2 (progress tersimpan, tapi resume per-pertanyaan di tengah jalan belum)
- ☐ Latihan soal L1: mulai dari soal 1 → reward→ level 2 dsb (starter set dipakai ulang utk L2/L3 sementara)

1.27.5 Fitur Tambahan & UX Polish — DONE

- ☒ Localization penuh: Indonesia, English, Basa Jawa (Krama Inggil) via i18next + ekstraksi data layer
- ☒ Audio engine Gamelan Web Audio API (Ombak, saron, gong ageng, keprak, detuned partials)
- ☒ Hide global scrollbars untuk PWA fullscreen feel
- ☒ Mengunci Orientasi Layar (Portrait PWA Lock)

1.27.6 PWA & offline

- ☒ Service worker (Workbox) precache penuh (~500 KB, 14 entries) — FR-6.2

- ☒ Update banner prompt-refresh — FR-6.3 (registerSW onNeedRefresh, diperbarui ke autoUpdate)
- ☒ Uji playthrough offline penuh di perangkat (IOS icons + user-scalable=no)

1.27.7 QA rilis

- ☒ Unit test matcher/raster + geometry + dtw (8 tests) — vitest
- ☐ Coverage 90% utk engine (target paket tambahan)
- ☐ E2E Playwright: Home→L1 selesai→L2 unlock; reload mid-level
- ☐ Matrix perangkat (iPad, Android tablet, desktop) — §14

1.28 P1 — Penting (peluncuran berkualitas)

- ☒ Partikel sukses per soal & reward (60 particle) — §9.5 (Selesai via Anime.js confetti)
- ☒ Suara gamelan ringan (Web Audio) (engine audio.ts selesai) — §12.2
- ☒ Transisi antarmuka & animasi UI (menggunakan animejs dgn bounce/stagger)
- ☒ Transisi slide sinopsis (Prolog.tsx) — FR-2.2
- ☐ Modal congrats/fokus trap — §9.7
- ☒ Audit a11y axe (P0 kritikal/serius = 0, kontras AA) — §16
- ☒ Skip-link + landmark announcement + `:focus-visible` penuh — §16
- ☒ Suport keyboard (story slides, Enter/Space) — §16 (Selesai untuk Prolog, Practice, Phase2, LevelDone)
- ☐ Alternatif aksesibel untuk canvas (deskripsi soal lengkap) — §16
- ☐ Optimasi bundle 1,5 MB + font 600 KB (saat ini ±500 KB total precache) — §15
- ☐ Pola 60fps saat menulis dgn guide menyala (rAF + dpr scaling)— §15
- [~] Kanvas toggle panduan (Tampilkan/Sembunyikan Contoh) — selesai, verifikasi UX
- ☐ E2E offline emulation penuh (Playwright)

1.29 P2 — Peningkatan (setelah rilis inti stabil)

- ☐ TTS pengucapan fonem + pelafalan nama aksara (lokal, gabung dgn suara)
- ☒ Transliterasi teks → Aksara Jawa (masukan bebas) — Selesai via “Free Type”
- ☐ Set glyph lanjutan: Angka, Murda, Swara
- ☐ Peningkatan model matcher → classifier ML (ONNX/TFLite <500 KB) — §11.3
- ☐ Statistik per latihan tambahan (skor, waktu, retry)
- ☐ Kuis pengetahuan cepat (bukan hanya menulis) sbg penguat
- ☐ Mode bantuan ekstra anak kecil (petunjuk lebih besar, warna kontras tinggi)
- ☐ Pengaturan lanjutan (volume, mode kontras tinggi, ukuran teks)

1.30 P3 — Masa depan (di luar MVP)

- ☒ Dockerization untuk hosting production yang mandiri
- ☐ Cloud sync & laporan guru (“Guru Report”) — §19
- ☐ Leaderboard lokal/kelas + akun opsional — G5, §19

- ☐ APK Android via TWA; wrapper iPad — §19
- ☒ Multi-bahasa (Jawa–Indonesia–English) — Selesai (i18n ditambahkan ke MVP)
- ☐ Editor stroke konten untuk guru (impor data aksara baru)
- ☐ Ekspor sertifikat penyelesaian (printer-friendly)

1.31 Notulensi Pengunci Keputusan

#	Keputusan	Status	Deadlines
1	Jumlah soal Level 1 (starter: 5 glyph ha/na/ca/ra/ka)	terbuka	sebelum isi dataset penuh
2	Isi dataset (glyph mana saja per level)	terbuka	sebelum dataset lengkap
3	Sinopsis L1 (copy lengkap dari PDF hilang)	terbuka	sebelum copy cerita penuh
4	Framework React vs Svelte	**React** (diputuskan)	—
5	Aset ilustrasi (SVG inline)	terbuka	akhir minggu 2
6	Model evaluasi menulis	**Raster coverage** (DTW outline diganti)	—
7	Sumber data pen-stroke kurasi (guide internal)	terbuka	P1

Perbarui notulensi § berdasarkan keputusan TIMELINE.md; kotak cek dicoret saat selesai.